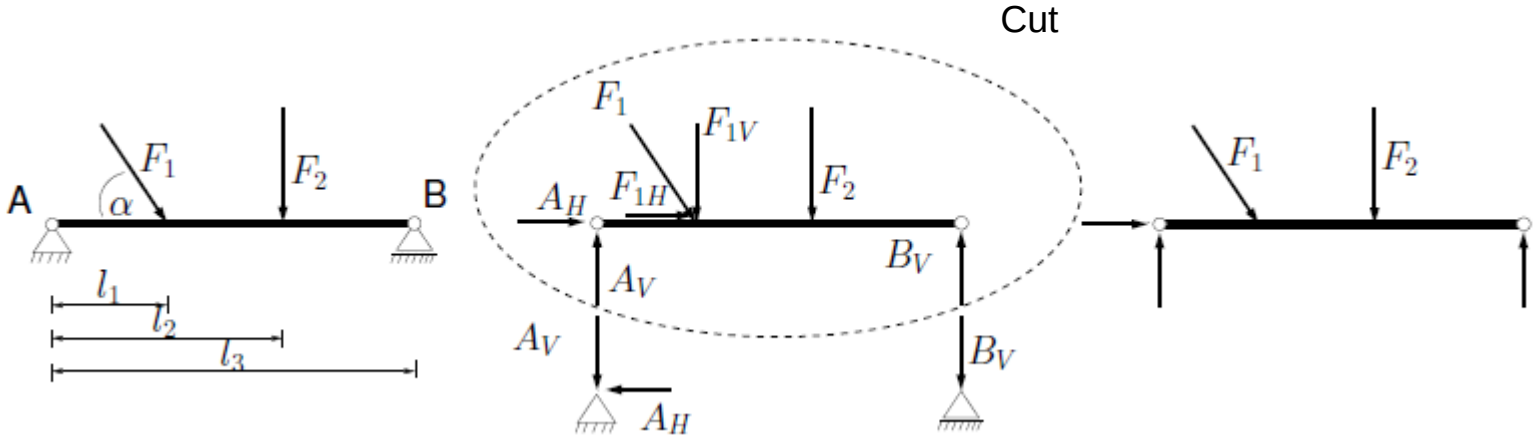
5.5. Calculation of support and intermediate reactions

Procedure:

- Cut supports and joints free
- Apply associated force actions with orientation
- Calculate all forces from equilibrium conditions
- Actual force direction from sign of result
    - Positive sign: Force acts in the applied direction
    - Negative sign: Force acts against the applied direction

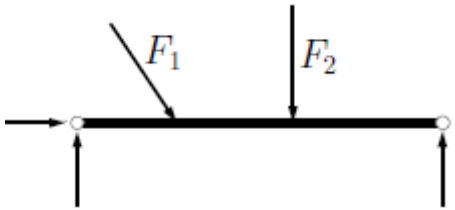
5.5.2. Example: Single span beam



$$F_{1V} = F_1 \sin \alpha$$
$$F_{1H} = F_1 \cos \alpha$$

Wanted: Support reactions

5.5.2. Example: Single span beam



- From the equilibrium conditions, the support reactions can be calculated as follows:

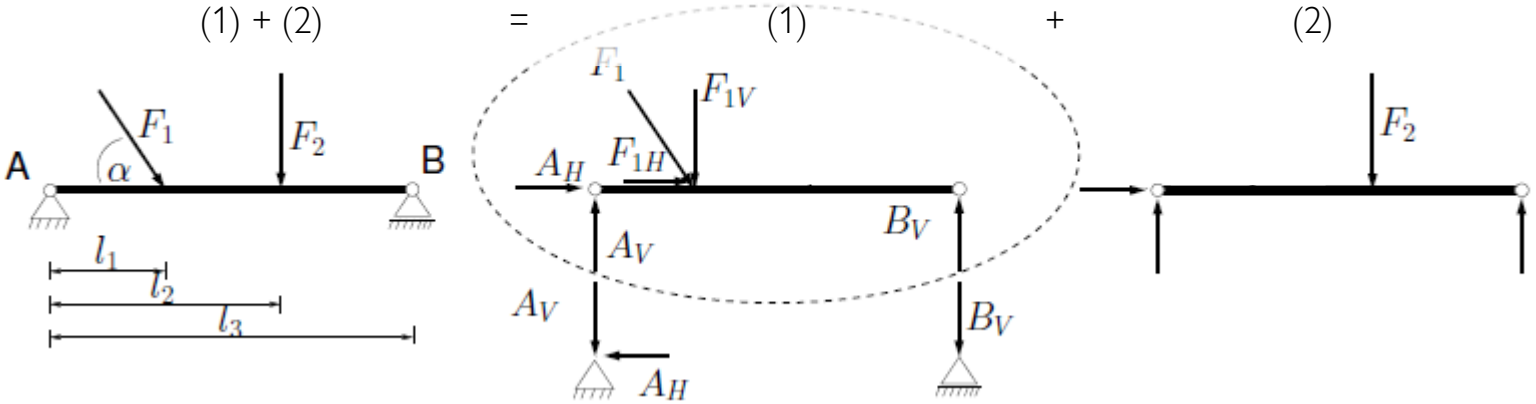
Box 5.14

5.5.2. Example: Single field beam

- Check:

$$\circlearrowleft \sum M_B = 0 : A_V \cdot l_3 - F_{1V} \cdot (l_3 - l_1) - F_2 \cdot (l_3 - l_2) = 0$$

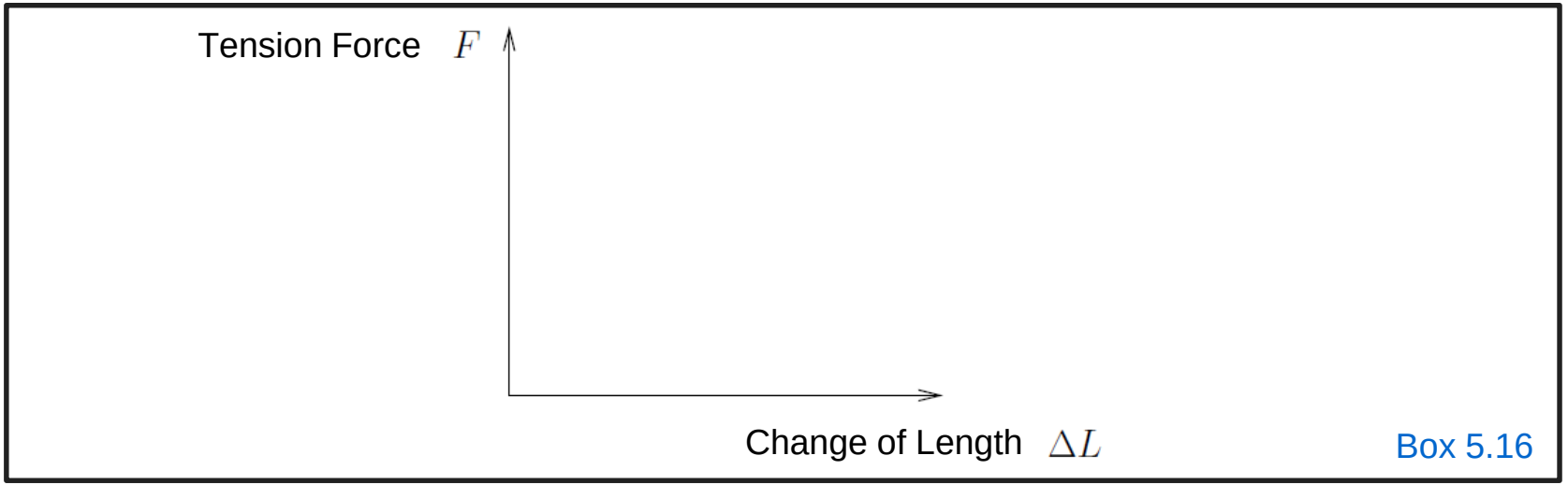
5.5.3. Superposition principle on the single field beam example



Box 5.15

5.5.3. Superposition principle on the single field beam example

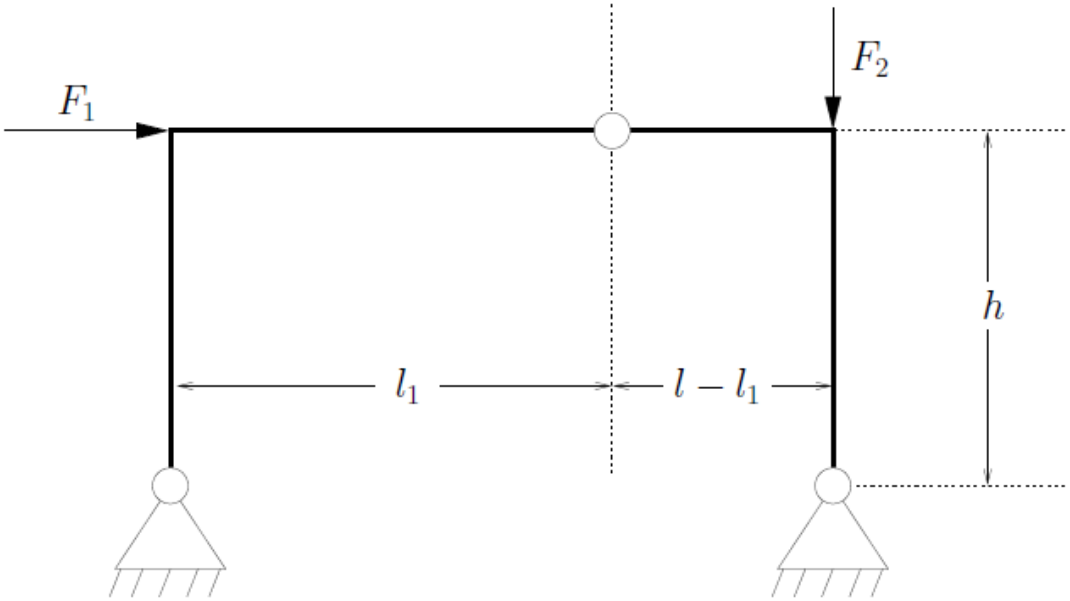
- Remarks



Box 5.16

The superposition principle is **not valid** for nonlinear theories!

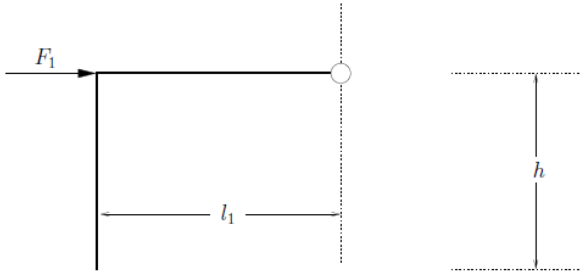
5.5.4. The three-hinged frame as a further calculation example



Wanted: Support reactions

5.5.4. The three-hinged frame as a further calculation example

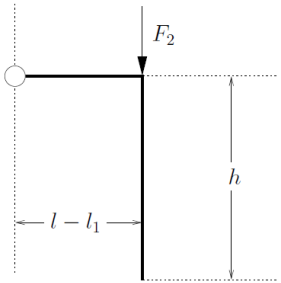
- Schematic procedure: left part of the structure



Box 5.17

5.5.4. The three-hinged frame as a further calculation example

- Schematic procedure: right part of the structure



Box 5.18

5.5.4. The three-hinged frame as a further calculation example

- 6 Equations with 6 unknowns
- Solving results in:

$$B_V = F_1 \cdot \frac{h}{l} + F_2$$

$$A_V = -\frac{h}{l} \cdot F_1$$

$$A_H = -\frac{l_1}{l} \cdot F_1$$

$$B_H = \frac{l - l_1}{l} \cdot F_1$$

5. General plane structures - Quiz

- Which of the following statements is correct?

  a) An immovable system is always statically determined.
  b) An unmovable system is always kinematically determinate.
  c) The counting criterion helps in the evaluation of static determinacy.
  d) The counting criterion guarantees static determinacy.
  e) The pole criterion guarantees kinematic determinacy.
  f) The construction of a pole plan is influenced by the load.